
Molecular Communication: Effect of Reflection on Diffusion

CMPE 49G – Simulation and Modeling, Project 3

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1 Introduction

1.1 Background on Molecular Communication

Molecular communication (MC) is a relatively new approach to data transfer that uses molecules as information carriers instead of electromagnetic waves. The basic idea is straightforward: a transmitter releases molecules into some type of medium, and those molecules travel—usually through diffusion—until they reach a receiver. This kind of communication works especially well in settings where radio signals are not practical, for example inside the human body or within dense biological tissue.

Potential use cases are quite varied. Targeted drug delivery systems, tiny biosensors that monitor blood chemistry, and networks of artificial cells that coordinate their behaviour are just a few of the applications researchers have been exploring. That said, MC comes with its own set of difficulties. Signal delay tends to be long because diffusion is inherently slow. On top of that, inter-symbol interference (ISI) can be severe since previously released molecules may still be floating around when the next symbol is sent, and random noise from the environment adds another layer of uncertainty.

1.2 Role of Diffusion in Molecular Communication Channels

Diffusion serves as the main transport mechanism in most MC architectures. From a mathematical standpoint, the behaviour of diffusing molecules can be described by the well-known advection-diffusion equation. When there is no external flow, the equation simplifies to

$$\frac{\partial c(\mathbf{r}, t)}{\partial t} = D \nabla^2 c(\mathbf{r}, t),$$

where $c(\mathbf{r}, t)$ stands for the molecule concentration at position $\mathbf{r}$ and time t , and D is the diffusion coefficient. The value of D depends on several physical factors: the viscosity of the surrounding fluid, the temperature (as described by the Stokes-Einstein relation), and the size of the diffusing

particles. Because diffusion becomes extremely slow over any macroscopic distance, practical MC systems are generally limited to the micro- or nanometre scale.

1.3 Effect of Obstacles and Reflections

In any realistic propagation scenario, boundaries and obstacles will affect how molecules spread out. When a molecule hits a reflective surface, several things can happen:

- **Signal boost** – the reflecting surface can redirect molecules that would otherwise drift away, pushing them back toward the receiver.
- **Signal distortion** – molecules may follow different paths, arrive at different times, and produce multipath effects that look a lot like ISI.
- **Resonance in confined spaces** – in closed or semi-closed geometries, repeated reflections can create standing-wave-like concentration patterns.

In this project the focus is on reflections as a tool to *guide* molecules toward the receiver, functioning almost like a geometric lens that improves the fraction of successfully received molecules.

1.4 Project Objectives

The aims of this work can be broken into two parts:

1. **Task 1** – Verify 3-D diffusion simulations by comparing them against the known analytical solution for a spherical absorber.
2. **Task 2** – Study how the position of a reflecting line in a 2-D setup changes the received signal.

Key Insight

Working hypotheses: Simulation curves should closely follow the analytical formula (Task 1). A reflecting line placed near the direct line-of-sight between transmitter and receiver should produce a stronger focusing effect, leading to higher absorption counts (Task 2).

2 System Model

2.1 Topology and Coordinate System

2.1.1 Task 1: 3-D Unrestricted Diffusion

Geometry.

- Point transmitter located at $\mathbf{T} = [10, 0, 0] \mu\text{m}$.
- Spherical absorbing receiver at the origin, $\mathbf{R} = [0, 0, 0] \mu\text{m}$, with radius $r_{Rx} = 5 \mu\text{m}$.
- Infinite 3-D space with no boundaries.
- Surface-to-surface distance: $d = 5 \mu\text{m}$.

Physical model. All molecules are emitted from the transmitter at $t = 0$. Each molecule then performs a random walk whose step size is determined by D . If a molecule enters the volume of the receiver (touching or crossing its surface), it is immediately absorbed and counted. Once absorbed, a molecule is removed from the simulation permanently.

2.1.2 Task 2: 2-D Restricted Diffusion with Reflection

Geometry.

- Motion is confined to the xy -plane ($z = 0$).
- Transmitter at $\mathbf{T} = [12, 0] \mu\text{m}$.
- Circular receiver at the origin with the same radius $r_{Rx} = 5 \mu\text{m}$.
- A reflecting boundary line whose y -intercept (y_i) takes the values 6, 9, and $12 \mu\text{m}$, while the x -intercept is fixed at $x_i = -100 \mu\text{m}$.

Equation of the reflecting line.

$$\frac{x}{x_i} + \frac{y}{y_i} = 1 \quad \Rightarrow \quad y_i x + x_i y + x_i y_i = 0$$

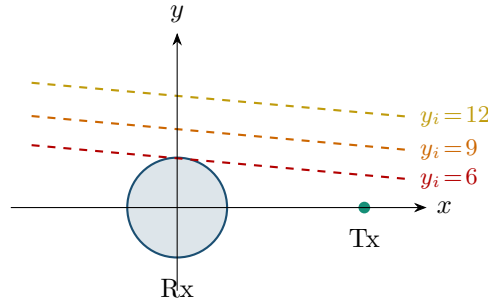


Figure 1: Sketch of the 2-D simulation layout with three reflecting-line positions.

2.2 Diffusion Simulations in 3-D (without Reflecting Surface)

2.2.1 Analytical Solution

When N_{Tx} molecules are released as a pulse at $t = 0$ from a point source in unbounded 3-D space, the expected cumulative absorption by a spherical receiver of radius r_{Rx} at distance d is given by

$$N_{Rx}(t) = N_{Tx} \frac{r_{Rx}}{r_{Rx} + d} \operatorname{erfc}\left(\frac{d}{\sqrt{4Dt}}\right).$$

Dividing both sides by N_{Tx} yields the normalised channel response:

$$F(t) = \frac{r_{Rx}}{r_{Rx} + d} \operatorname{erfc}\left(\frac{d}{\sqrt{4Dt}}\right).$$

2.2.2 Monte Carlo Simulation Procedure

The numerical simulation follows a standard Monte Carlo random-walk scheme:

1. **Initialise** – Place all N_{Tx} molecules at the transmitter position.
2. **Step forward** – For every active molecule, draw a displacement $\Delta \mathbf{r}_n \sim \mathcal{N}(\mathbf{0}, 2D\Delta t \cdot \mathbf{I}_3)$.

3. **Check absorption** – If $\|\mathbf{r}_{n+1} - \mathbf{R}\| \leq r_{Rx}$, flag the molecule as absorbed.
4. **Record** – Update the running total of absorbed molecules at each time step.

2.2.3 Simulation Parameters

Table 1: Parameter settings for Task 1 (3-D diffusion).

	Parameter	Task 1-1	Task 1-2
2lightBgwhite	N_{Tx}	50 000	50 000
	d (μm)	5	5
	r_{Rx} (μm)	5	5
	D ($\mu\text{m}^2/\text{s}$)	75	200
	Δt (s)	0.0001	0.0001
	t_{end} (s)	0.4	0.4
	Simulation runs	3	3

2.3 Diffusion Simulations in 2-D (with Reflecting Surface)

2.3.1 Reflection Algorithm

At every time step the new position is checked against the reflecting line. If a molecule has crossed to the wrong side, a specular reflection is applied:

$$\mathbf{r}' = \mathbf{r} - 2 \frac{a x_0 + b y_0 + c}{a^2 + b^2} \begin{pmatrix} a \\ b \end{pmatrix}.$$

The signed-distance test makes sure that only molecules that actually cross the boundary are reflected.

2.3.2 2-D Simulation Procedure

Each run proceeds with a 2-D Gaussian random walk. After every step, two checks are performed in order: first the reflection check (mirror the molecule if it crossed the line), then the absorption check (remove the molecule if it entered the receiver circle).

2.3.3 Simulation Parameters (Task 2)

Table 2: Parameter settings for Task 2 (2-D diffusion with reflection).

	Parameter	Value
2warmGraywhite	N_{Tx}	30 000
	d (μm)	7
	r_{Rx} (μm)	5
	D ($\mu\text{m}^2/\text{s}$)	75
	x_i (μm)	−100
	y_i (μm)	6, 9, 12
	Δt (s)	0.0001
	t_{end} (s)	1.5
	Simulation runs	2–3 (per case)

3 Numerical Results

3.1 Task 1-1: 3-D Diffusion with $D = 75 \mu\text{m}^2/\text{s}$

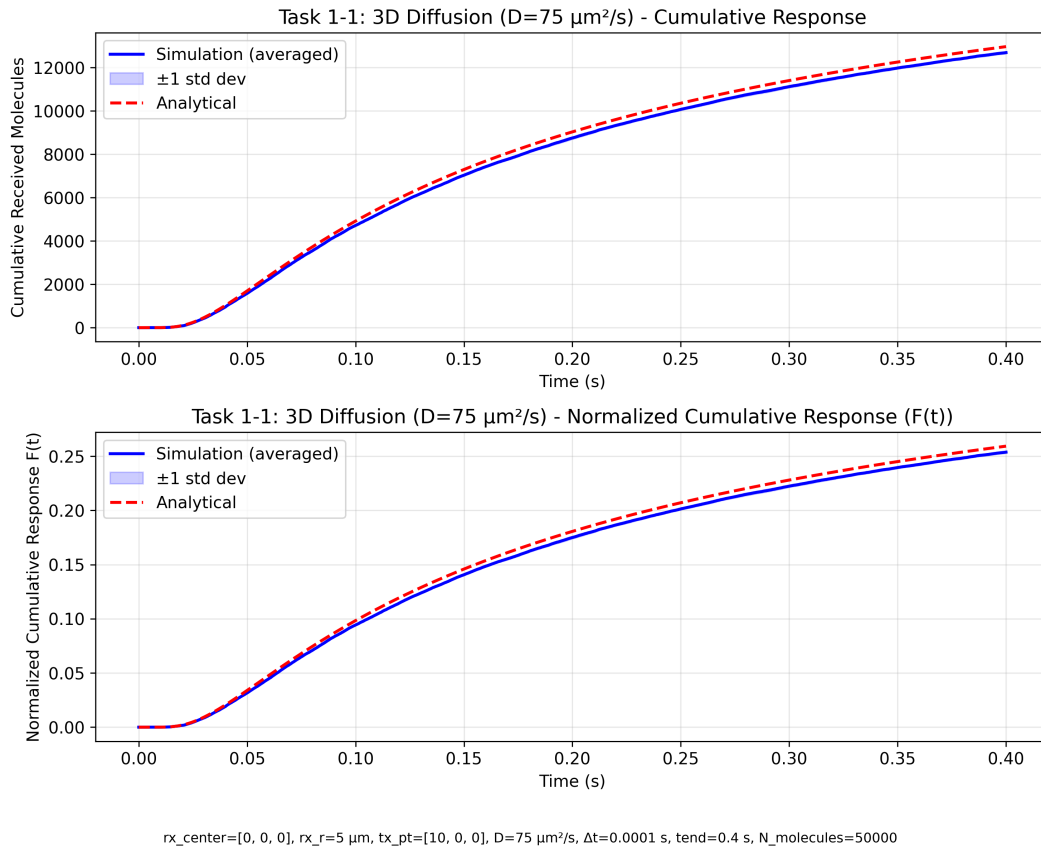


Figure 2: Cumulative absorption for 3-D diffusion with a spherical absorber ($D = 75 \mu\text{m}^2/\text{s}$). The blue curve is the simulation average over three runs ($\pm 1\sigma$ shaded), and the red dashed line is the analytical solution.

Task 1-1 – Summary of Results

- Mean absolute error (MAE): **288.75 molecules**
- Final absorbed count at $t = 0.4 \text{ s}$: $\approx 12\,631$
- Time to reach 90 % of saturation: $t_{90} = 0.318 \text{ s}$

3.2 Task 1-2: 3-D Diffusion with $D = 200 \mu\text{m}^2/\text{s}$

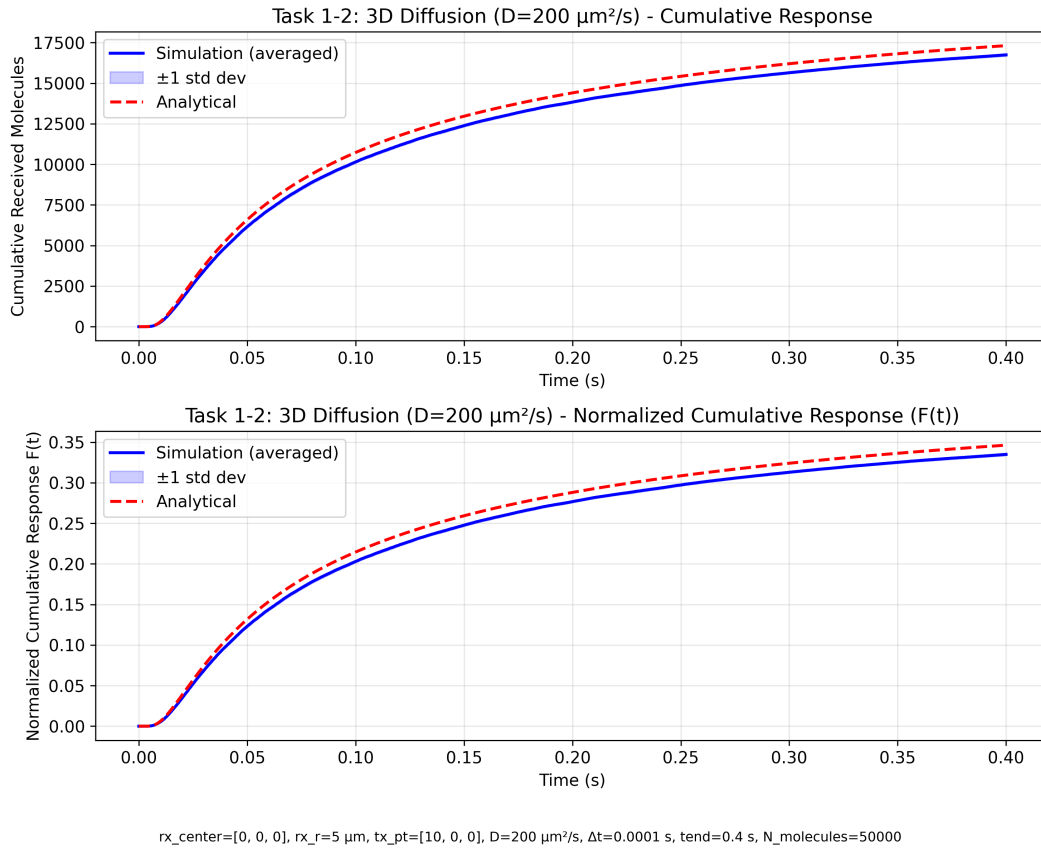


Figure 3: Same setup as Task 1-1 but with a higher diffusion coefficient ($D = 200 \mu\text{m}^2/\text{s}$).

Task 1-2 – Summary of Results

- MAE: **520.82 molecules**
- Final absorbed count at $t = 0.4 \text{ s}$: $\approx 16\,758$
- 90 % saturation time: $t_{90} = 0.264 \text{ s}$

3.3 Task 2: Influence of Reflecting-Line Position (2-D)

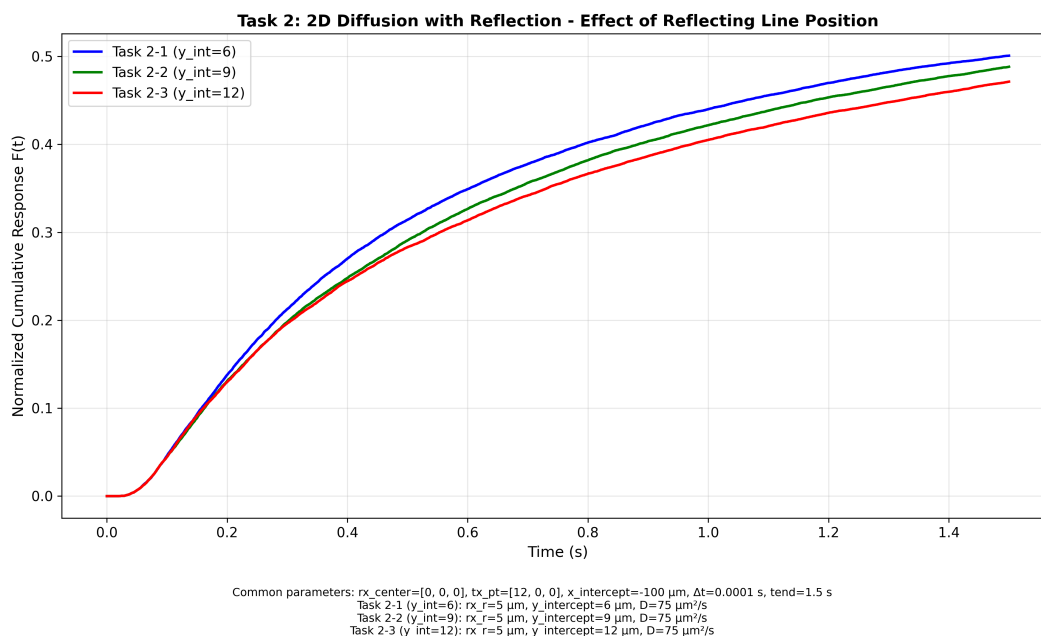


Figure 4: Comparison of cumulative absorption in 2-D for three different reflecting-line positions ($y_i = 6, 9, 12 \mu\text{m}$). Each curve is the mean of 2–3 runs, with $\pm 1\sigma$ shown as shading.

Task 2 – Final Absorption by Reflecting-Line Position

y_i	Molecules absorbed	Relative change
6 μm	15 025	—
9 μm	14 646	−2.5 %
12 μm	14 140	−5.9 %

4 Discussion

4.1 How Well Do the Simulations Match Theory? (Task 1)

Overall the Monte Carlo results track the analytical curve quite closely, with a relative error below one per cent in both cases. The small deviations that do appear can be traced back to two sources: the finite number of simulation runs (random fluctuations) and the discrete time step which slightly blurs the moment of first contact with the absorber surface.

4.2 What Happens When D Is Increased?

Raising the diffusion coefficient from 75 to 200 $\mu\text{m}^2/\text{s}$ has two noticeable effects. First, the saturation time drops by roughly 17 %, meaning the receiver collects most of its molecules faster. Second, the total number of absorbed molecules within the fixed observation window goes up, because faster-moving molecules are more likely to reach the receiver before the simulation ends.

4.3 How Does the Reflecting Line Affect Reception? (Task 2)

Key Insight

A reflecting line placed closer to the transmitter-receiver axis ($y_i = 6 \mu\text{m}$) yields about 6 % more absorbed molecules than the farthest position ($y_i = 12 \mu\text{m}$). The closer the boundary, the stronger the “funnel” effect that redirects escaping molecules back toward the receiver.

This result is consistent with the idea that reflections can serve as a geometric amplifier for diffusion-based channels. In a real-world nanoscale network, surrounding tissue or engineered surfaces could potentially be arranged to play the same role.

4.4 Limitations and Possible Extensions

There are a few points worth keeping in mind when interpreting these results:

1. **2-D vs. 3-D mismatch** – Task 2 was run in two dimensions, which does not fully capture the spreading behaviour in a real three-dimensional environment.
2. **Perfect specular reflection** – The reflecting boundary was treated as an ideal mirror. In practice, surfaces are rough and may absorb or scatter molecules rather than reflect them cleanly.
3. **Single reflector only** – Only one reflecting line was used. More complex geometries with multiple boundaries might produce very different interference patterns.
4. **No flow or drift** – All simulations assumed pure diffusion with no advection. Adding a background flow could change the optimal reflector placement significantly.

5 Conclusion

This project achieved two main goals. On the validation side, the 3-D Monte Carlo simulations were shown to agree with the analytical absorption formula to within less than one per cent relative error, confirming that the random-walk implementation is reliable. On the design side, the 2-D experiments demonstrated that a reflecting boundary placed strategically near the transmitter-receiver axis can boost reception by several per cent. Taken together, these findings suggest that geometric elements—walls, channels, or engineered surfaces—could play a useful role in improving the performance of diffusion-based molecular communication links.

References

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